SANGBAY
Gas Sensor

S4-H2S sensor

S4-H2S Application Specification

Electrochemical Hydrogen Sulfide Sensor

Small size | Low power consumption | Long life | Low cost | High precision

Shanghai Sangbay Sensor Technology Co., Ltd.

1. Product Overview

The S4-H2S hydrogen sulfide gas detection sensor is a constant-potential electrolytic sensor. Hydrogen sulfide and oxygen undergo corresponding redox reactions on the working electrode and the counter electrode, respectively, and release charges to form a current. The generated current is proportional to the concentration of hydrogen sulfide. The hydrogen sulfide concentration can be determined by testing the current.



2. Application Field

① Hydrogen sulfide alarm

② Industrial site hydrogen sulfide gas detection

③ Mine hydrogen sulfide detection

④ Hydrogen sulfide detection in atmospheric environment

3. Product Characteristics

① High precision and long life

② Fast response speed, fast return to zero

③ Low power consumption, high sensitivity

④ Wide linear range, strong anti-interference ability

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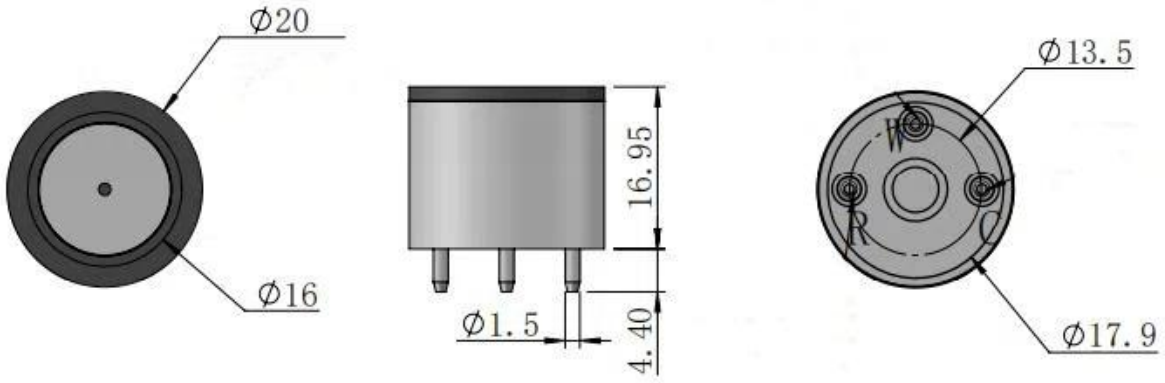
⑤ Excellent repeatability and stability

⑥ Strong anti-interference ability

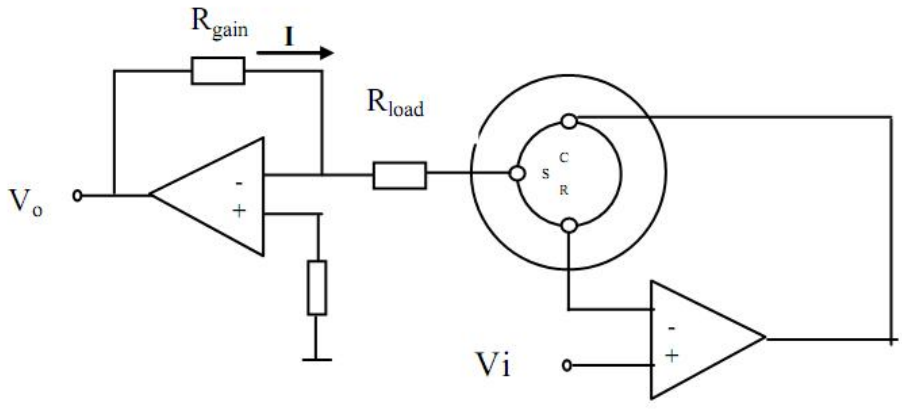
4. Technical Parameters

Item	Parameter
model	S4-H2S-100
Detection principle	Electrochemical Three electrodes
Detection gas	H2S
Detection range	0-100PPM
Maximum load concentration	500PPM
Sensitivity	650±200nA/PPM
Zero drift	±2PPM
Resolution	1PPM
Response time	< 30s
Bias voltage	0
Load resistance	5~30Ω
Load resistance	-30℃-50℃
Load resistance	15%RH to 90%RH(no condensation)
Repeatability	1% output signal
Long-term stability	< 2% signal/month
Linearity	Linearity,regression coefficient R^2=0.999
Work pressure	90 to 110 kPa

Shelf life	12 months after delivery
Service life	2 years

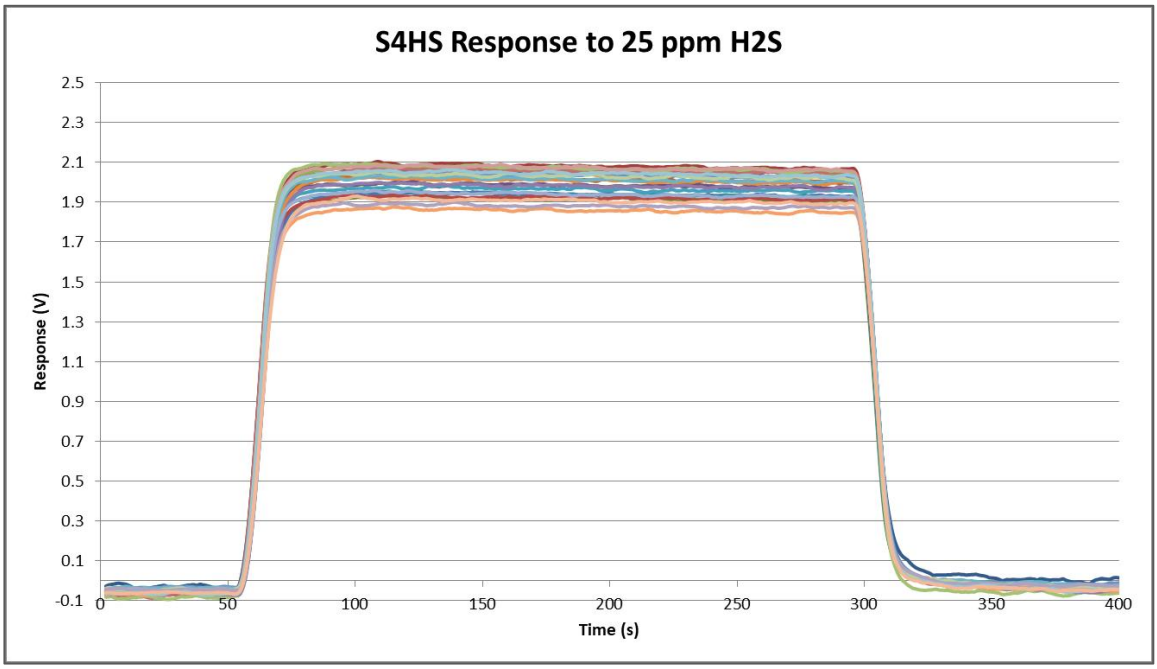


5. Basic Circuit

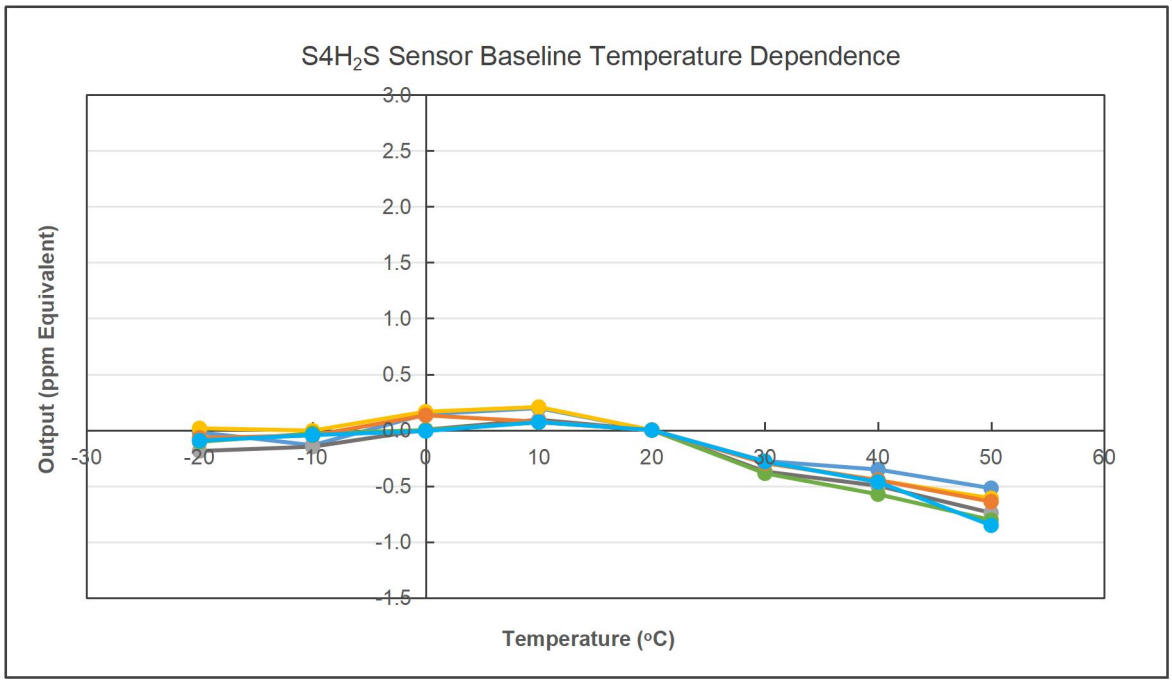


Note: The above picture shows the basic test circuit of the S4-H2S sensor.

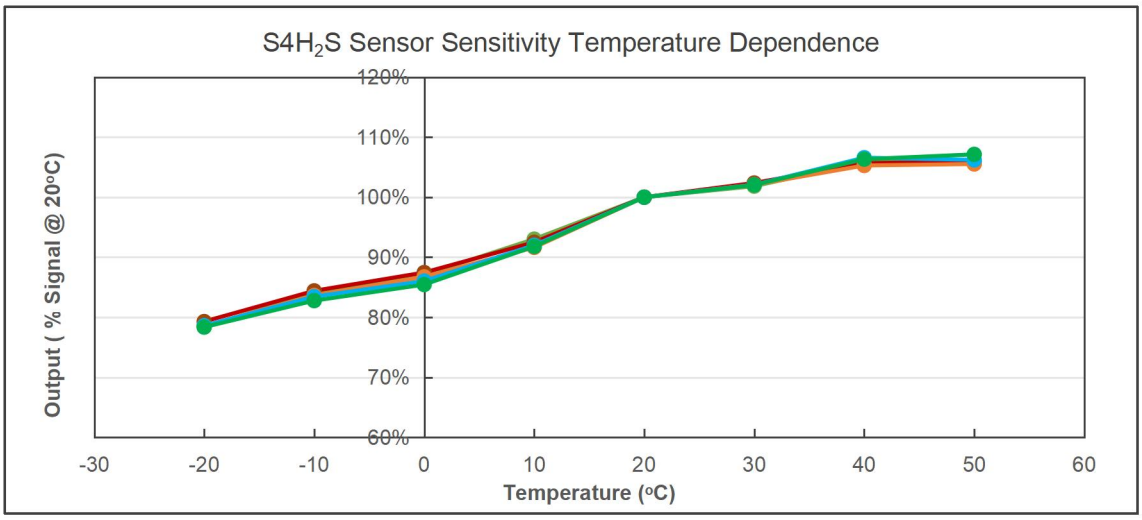
6. Sensor Characteristic Description



Sensor response recovery curve



Sensor typical sensitivity characteristic curve



Sensor temperature and humidity characteristic curve

7. Cross Interference:

The S4-H2S sensor also responds to gases other than the target gas. The response characteristics of the sensor to several common interference gases are listed in the table below for reference. The data in the table is the typical response of interfering gas at a given concentration.

Interfering gas	used gas concentration (ppm)	isplay value (ppm H_2S)
CO	300	1
SO_2	20	1
NH_3	50	0
NO_2	20	-1
H_2	1000	0
NO	50	2
CL_2	20	-1.4